

FSC 203: Science, Technology and Society

1. What is Science; What is Technology. Relationship between Science and Technology.
2. The Role of Science and Technology in the Society.

1. MEANING OF SCIENCE

Science can be defined as a process of investigation involving activities or processes known as science skills; which include observation, hypothesizing, measuring, experimenting and so on. These activities lead to the production of body of knowledge which are the product of science. This body of knowledge comprises of principles, facts, theories, concepts and laws. Science can be defined as a systematized body of knowledge and a process of inquiry carried out in order to study the world around us. Science is an intellectual activity through which men seeks to understand nature. Science is an intellectual activity carried out by human that are designed to discover information about the natural world in which humans live and to discover the ways in which this information can be organized into meaningful patterns.

Moreover, science is a system developing a scientifically based knowledge base on defined method of inquiry, as well as to the organized body of knowledge humans have achieved by such research. Science refers to a systematic venture that shape and establishes knowledge in the form of verifiable predictions and explanation on global developments. It is the compendium of reliable ideas that can be rationally and logically explained. Science involves investigations of the physical world, including us and the stuff we make. • Science is a formal activity that creates knowledge by direct interaction with nature. • Science has some kind of special method that allows different scientists to produce the same kind of knowledge whatever their social and political context might be. • Scientists perform the same experiments in the same way, and agree upon and reject the same hypotheses. • Scientists come to consensus on the truths of the natural world. According to this view, when we do science, we set aside values and study only facts.

The product content of science refers to the organized body of knowledge embedded in facts, concepts, principles, laws, and theories. It is the product of scientific investigations. It is through the use of the products of science that regularities in nature are described, explained and

predicted. This organized body of knowledge is testable, falsifiable, verifiable, contingent and debatable.

Science is a process and means by which man investigates his world and gains knowledge about it. The processes of science relate to those activities carried out by scientists during a scientific investigation include: • Observation • Classification • Measurement • Prediction • Problem identification • Testing hypothesis • Analyzing • Inferring • Experimenting • Collection of data • Interpretation of data • Extrapolating • Drawing valid conclusion etc.

From the above list, it is evident that scientific processes involve several activities in an effort to obtain valid generalization and to raise the right kinds of questions about objects and events around us.

Mani (1985) defined science as a systematized body of knowledge and a process of inquiry carried out in order to study the world around us.

Holbrook (1992) defined science as an intellectual activity through which men seek to understand nature.

Gottlieb (2004) viewed science as an intellectual activity carried out by human that are designed to discover information about the natural world in which human

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What is Technology

Technology as a disciplined process that uses the resources of matter, energy and natural phenomena to achieve human purpose. It is the practical application of scientific results for the development of tools, equipment and techniques. Thus, the invention of the telescope, of techniques of cooking, canning, bottling or preservation, or of medicaments to reduce body pains are seen as technologies. Technology employs knowledge, skills and tools to improve human potentials, to solve practical problems, to modify our environment. It is therefore concerned with the application of science to obtain practical solution to the myriad of human problems.

Technology simply put is a means of harnessing and exploiting our understanding of nature for our own benefit. It is an application of knowledge for practical purpose. It is used to improve human condition, natural environment or to carry out other socio-economic activities. It could also be defined or refers to all processes dealing with materials and their end products. One important attribute of technology is that it does not just happen; it is developed and learned whether in the form of manual skills or as an applied science. It is the systematic application of collective human rationality to the solution of human problems through the assertion of control over nature; technology is the engine of growth. Technology can be traced historically to the beginning of time to be man's quest to improve his way and quality of life.

Technology is the total and complete application of man's knowledge, skills, tools and materials. It is the use of scientific knowledge to develop and produce goods and services useful to man. It is practical problem-solving enterprise, which is propelled by scientific discovery or by societal needs. Technology therefore can be broadly classified into two major categories namely: (a) Material Technology- where knowledge is embedded into technological products such as tools, equipment, agro-chemicals, improved plant varieties or hybrids, improved breeds of animals and vaccines. (b) Knowledge-Based Technology- such as technical knowledge, management skills and other processes which are needed to successfully produce products or grow crops.

Society

Society can be defined as a group of humans living together for self maintenance and self-perpetuating and sharing their own institution and culture. It could also be defined as a long-standing group of people sharing cultural aspects such as languages, dress, norms of behaviours and artistic forms. A formal association of people with similar interest

Role of Science and Technology in Agriculture Advance in science and technology have facilitated higher yields, greater efficiency and greater nutritional content in the world's food supply. Food production, however, must double in the coming decades to meet rising demand and meet the challenges entailed in inter alia improving resistance to drought, pests, salinity and temperature extremes, raising the nutritional content and reducing post-harvest loss all in an environmentally and socially sustainable manner. At the same time, the regulatory challenges of assuring safety in food production and consumption will increase exponentially. Although, the necessary

technology for increasing the agricultural output is available in many developing countries including Nigeria but certain requisites must be met e.g Medical Society, Science Teachers Society are all examples of societies.

Module 2: The role of Science and Technology in Society.

Science and technology make life easier and more comfortable for humans. Science and technology enable us to save both time and money. Science and technology open up new perspectives of understanding

2.1 Role of Science and Technology in Societal Development

Science and technology hold the key to the present and future development of Nigeria or any other country for that matter. Technology plays a fundamental role in wealth creation, improvement of the quality of life and real economic growth and transformation in any society. Egbogah (2012) maintained that the sooner Nigeria realizes that her escape from poverty is predicated on her investment in science and technology education, the better for her. Animalu (2001) in Egbogah (2012) explained that there is a technological power vacuum in Nigeria waiting to be filled by which ever geo-political zone that cares to mobilize its people through dedicated and selfless services. Essentially, technology is the primary engine of economic growth. It is the key and fundamental requirement for value addition to raw materials and people. It provides the key to unlocking any country's potential in terms of decreasing over-head costs associated with out sourcing and creating employment opportunities. Science and technology education will not only prepare the Youths of Nigeria and indeed any other nation, for fulfilling career prospects, but also train their minds to address social problems with scientific mind. Youths equipped with science and technology education are also endowed with high employment opportunities. Many developed and advanced countries did progress much because of their heavy investments on science and technology. Examples: The United Kingdom and France benefited immensely from the industrial revolution in the 19th century. Similarly, the United States emerged from an agrarian economy in the 19th century into an industrial superpower in the 20th century. More recently, Taiwan and Korea have exploited advances in silicon microelectronics from the early 1960s. China and India have emerged as industrial leaders in manufacturing and

information technology respectively. Malaysia has also followed in the footsteps of these later Asian successes. It is necessary to emphasize here that in recorded achievements all these countries invested heavily in people, factories and infrastructure that provided the foundation for today's industries. These successes were all based on carefully designed roadmaps of plans and strategies. Unfortunately, however in many if not all the non-development or yet to develop countries, technology is seen or viewed as a consumable- item, and not something that can be produced or created.

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Role of Science and Technology in Poverty Alleviation

Science and technology have been central in the progress made to date in the fight against poverty and in stimulating economic growth. Advances in science and technology are, in many ways, the ultimate Global Public Good: once discovered, their benefits can be extended to additional users at little or no marginal costs. In most basic and critical areas of human need, science and technology have made possible significant progress to date, and they hold the best prospects for continued progress, particularly with respect to agriculture, health, energy, water, and environmental concerns.

Role of Science and Technology in Health

The most important contribution of science and technology to health is the ability to provide cures or prevention measures for the different ailments that afflict man. Some of the contributions of science and technology to health includes;

- Discovery of complex techniques to cure previously incurable diseases
 - Development of vaccines to prevent attack by some diseases like measles, small pox polio, yellow fever
 - Complex surgical equipment to make previously complicated surgeries simple
 - Medication for all types of diseases
 - Diagnostic equipment that help in detecting diseases
- X- rays, machine, scanning machine
- Development of genetic engineering
 - Biotechnology industry.

Role of Science and Technology in Communication

Science and technology have influenced the means of communication. In the olden days messages took months to reach distance places. Messengers and horse – riders carried messages. Now there are numerous means of communication like telephones, wireless, radios, televisions, and so on. On a telephone you can talk to your friends, relatives at a distant place in few seconds. With the help of wireless apparatus, messages may be conveyed to far off places in no time in fact. telecommunication has made the whole world a “global village”. Radios give us news from all the corners of the world, while on the television screen you can see happenings of far-off lands.

Role of Science and Technology in transportation

Science and technology have revolutionized the area of transportation. In the olden days means of transportation were by trekking, and horse riding. With advent of science and technology travelling by land, using cars, buses, trains; by sea using ships and canoes, by air using aero planes is now very easy. Aero planes are the fastest means of transportation. Time and distance are no more a problem to the society because of the means of transportation. The world has grown smaller. The credit for all these achievements goes to science.

Role of Science and Technology in Appliances in The Homes

Science and technology have helped in the production of appliances in the homes which make life more comfortable. Some of these appliances are: • Freezers and refrigerator which are used in the preservation of food as well as cooling drinks • Fans, air condition used during hot season • Room- heaters used when there is cold • Micro waves • Pressure cooker

• Washing machine • Pressing iron • Cooking pots • Electric kettle • Electric cooker • Gas cooker • Kerosine stove